

Comment on the Green's function for the anharmonic oscillators

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Application of the analytic theory of continued fractions to anharmonic oscillators of the type $ax^2 + bx^4 + cx^6$ by Singh *et al.* is shown to give an unphysical result for negative coupling, $b < 0$. For $b > 0$, their construction of the Green's function $G(E)$ is rigorously proved and extended to the $b \leq 0$ cases.

Let us consider the Schrödinger equation

$$-\frac{d^2}{dx^2}\psi(x) + V(x)\psi(x) = E\psi(x), \quad (1)$$

$$V(x) = ax^2 + bx^4 + cx^6, \quad c = \alpha^2 > 0.$$

In accord with Singh *et al.*,¹ we may write its general solutions $\psi_0(x)$, $\psi_1(x)$ in the standard power-series form

$$\psi_\nu(x) = x^\nu e^{-f(x)} \phi(x), \quad (2)$$

$$\phi(x) = \sum_{n=0}^N a_n x^{2n}, \quad \nu = 0, 1,$$

$$f(x) = \frac{1}{4}\alpha x^4 + \frac{1}{4\alpha}bx^2, \quad N \leq \infty,$$

where the a_n satisfy recurrence relations of the type

$$\sum_{m=n-1}^{n+1} Q_{nm} a_m = E a_n, \quad n = 0, 1, \dots \quad (3)$$

(see Ref. 1 for further details). The authors of Ref. 1 succeeded in finding an explicit ($n \times n$ -determinantal) form of the Taylor coefficients a_n . Requiring termination of the infinite series in Eq. (2), they also obtained an important class of exactly polynomial bound-state solutions $\psi(x)$ for some particular values of a , b , and c in Eq. (1). These results were successfully generalized recently.²

Our present aim is to make a few comments concerning the second half of Ref. 1—construction of the physical Green's function $G(E)$ for Eq. (1).

(i) The Hill-determinant method is not fully applicable to Eq. (1) in the formulation of Ref. 1: Indeed, the incorrect initial methodical statements¹ are the following:

(a) "The necessary and sufficient condition that nontrivial a_n 's (for $n = 0, 1, \dots$) exist which solve Eq. (3) is that the ... infinite determinant $\det(Q - E)$ vanish."

(b) "The difference equation [Eq. (3)] is an equivalent description of the original differential equation (1) for the eigenvalue problem."

Proof. Ad (a) Whenever $N = \infty$, we may choose the value of a_0 as an arbitrary normalization. Then,

the recursive treatment of Eq. (3) gives the explicit determinantal formulas for any a_n [$a_1 = -a_0 Q_{00}/Q_{01}$, $a_2 = a_0(Q_{00}Q_{11} - Q_{10}Q_{01})/Q_{01}Q_{12}$, etc.] as given by Eq. (11) in Ref. 1. In general, these a_n 's are nonzero irrespective of the value of E . Even for $N < \infty$, we must guarantee that $a_{N+1} = 0$ and $a_{N+2} = 0$. The first condition coincides indeed with the Hill-determinant specification of energies while the second one [$Q_{N+1N} = 0$, cf. Eq. (18) in Ref. 1] fixes one of the couplings as a function of N .

Ad (b) For $N = \infty$ and E not lying in the spectrum of Eq. (1) we still have¹ $a_{n+1}/a_n \sim O(1/\sqrt{n})$, $n \gg 1$ since $\phi(x)$ is convergent for any x (though not in the norm). Nevertheless, the recurrently defined vector $(a_0, a_1, \dots)$ remains normalizable ($\sum a_n^2 < \infty$) and must therefore be admitted as a valid solution to the eigenvalue problem Eq. (3) in principle. Q.E.D.

We may summarize as follows:

Ad (a) Instead of the "nontrivial" a_n 's in (a), we have to speak about their N -dependent "approximants" $a_n(N)$, and to replace the "infinite Hill determinant" in Eq. (9) of Ref. 1 by its N th minor with the N -dependent zeros $E(N)$, $N < \infty$.

Ad (b) When $N = \infty$ in Eq. (2), it is necessary to deliver the missing proof of coincidence of the physical eigenvalues of Eq. (1) with the intuitively chosen "approximants" $E(N)$ in the limit $N \rightarrow \infty$ (see below).

(ii) The algebraic manipulations of Ref. 1 only disguise the inapplicable Hill-determinant philosophy: Eq. (3) with $T_n = a_n/a_{n-1}$, $n = 1, 2, \dots$ is rewritten in the form¹

$$\begin{aligned} B_0 + A_0 T_1 &= 0, \\ C_1/T_1 + B_1 + A_1 T_2 &= 0, \\ C_2/T_2 + B_2 + A_2 T_3 &= 0, \\ &\dots \end{aligned} \quad (4)$$

where (a) the first row is omitted, (b) the continued-fraction approximants $T_n^{(N)}$, $n = N-1, N-2, \dots, 2, 1$ to the quantities T_n are defined by the initialization $T_N = T_N^{(N)} = 0$ at some $N \gg 1$, and (c) the first row is "self-consistently" included again in the limit $N \rightarrow \infty$.

In this setting, the “self-consistent eigenvalues” $E = E_0$, i.e., the zeros of the auxiliary continued fraction

$$\frac{1}{T_0^{(\infty)}} = -(B_0 + A_0 T_1^{(\infty)}) \quad (5)$$

are not derived from the Schrödinger boundary condition

$$|\psi(x)| < \infty \quad (6)$$

but from the purely formal requirement (b). As a consequence, it is possible to prove the following:

(iii) For $b < 0$, the Green's function of Singh *et al.*¹ defined as $G(E) = T_0^{(\infty)}(E)$ is unphysical:

(a) From the asymptotic form of Eq. (4)

$$T_n = \frac{\alpha}{-b/2\alpha + nT_{n+1}} \left[1 + O\left(\frac{1}{n}\right) \right] \quad (7)$$

we find out that the repeated action of mapping $T_{n+1}^{(N)} \rightarrow T_n^{(N)}$ leads in general to the accumulation of the values $T_n^{(N)}$, $N \gg n \gg 1$ near the N -independent fixed points,

$$T_n^{(N)} = Z_n \left[1 + O\left(\frac{1}{n}\right) \right], \quad Z_n = -\left(\frac{\alpha}{n}\right)^{1/2} \operatorname{sgn} b + O\left(\frac{1}{n}\right). \quad (8)$$

The proof follows from the geometrical interpretation of Eq. (7) as given in Fig. 1.

(b) The above asymptotic estimate of T_n and of $a_n = a_{n-1}T_n^{(\infty)} = a_{n-2}T_{n-1}^{(\infty)}T_n^{(\infty)} = \dots$ enables us to decompose

$$\begin{aligned} \phi(x) &= \phi_o(x) + \phi_e(x), \\ \phi_{o/e}(x) &= \sum_{n=\text{odd/even}}^{\infty} a_n x^{2n} = O(e^{2f(x)}), \quad x \gg 1 \end{aligned} \quad (9)$$

which follows from the Stirling formula. As a consequence, the quantity $\operatorname{sgn} T_n^{(\infty)}$, $n \gg 1$, determines the relative sign of the two growing exponents ϕ_o and ϕ_e . The asymptotic cancellation of ϕ_o with ϕ_e (i.e., $Z_n < 0$) is the necessary condition for the normalizability of $\psi(x)$ —this completes the proof.

Due to the rather exceptional character of the $x \rightarrow \infty$ boundary condition Eq. (6), a part of the construction of Singh *et al.* is correct:

(iv) For $b > 0$, the poles of $G(E) = T_0^{(\infty)}(E)$ coincide with the anharmonic binding energies: We may verify the algebraic identity

$$T_n = T_n^{(\infty)} + a_0 \prod_{j=1}^n \frac{C_j}{T_{j-1}^{(\infty)} A_{j-1}} = \frac{a_n}{a_{n-1}} \quad (10)$$

which is valid for any parameter E and values $1/T_{j-1}^{(\infty)} \neq 0$, $j = 1, 2, \dots, n$. At $E = E_0$, the difference $T_n - T_n^{(\infty)}$ [i.e., its $1/T_0^{(\infty)}(E)$ factor] changes sign. Since $T_n/T_n^{(\infty)} \sim 1 + \operatorname{const} \times (E - E_0) \sim \phi_o/\phi_e$

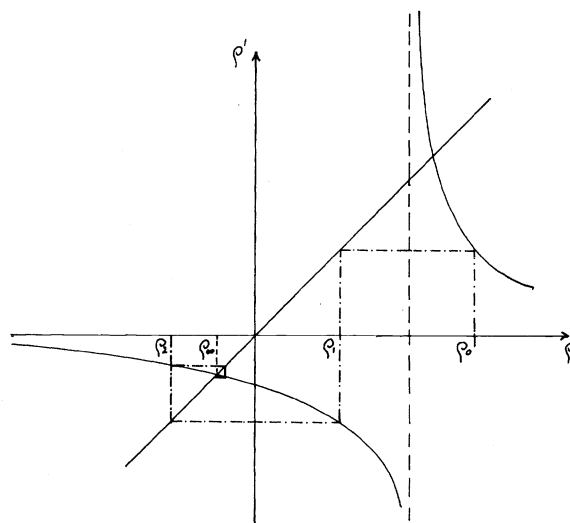


FIG. 1. The mapping $\rho = T_{n+1}^{(N)} \rightarrow \rho' = T_n^{(N)}$ with the fixed point $\rho_\infty = Z_n$.

for $x \gg 1$, the dominant part of $\phi_o + \phi_e$ also vanishes for $x \gg 1$. This represents the missing mathematical foundation of the construction of Singh *et al.* Comparison with (iii) clarifies also the puzzling discontinuity¹ of $T_0^{(\infty)}(E)$ at $b = 0$.

(v) We have seen that the physical fixed point

$$T_n = -(\alpha/n)^{1/2} + O(1/n) \quad (11)$$

of the mappings $T_n \leftrightarrow T_{n+1}$, $n \gg 1$ in Eq. (4) is determined uniquely by the cancellation requirement. For $b < 0$, it is stable if and only if the direction of the recurrences in Eq. (4) is reverted, $T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow \dots$, $T_1 = -B_0/A_0$. In full analogy with the construction of Singh *et al.*, the $b < 0$ oscillator may therefore be assigned the Green's function

$$G(E) = \frac{1}{N^{1/2} T_{N+1}(E)} \quad (12)$$

in the limit $N \rightarrow \infty$. In this formulation, only the initialization and termination roles of the first and N th ($N \rightarrow \infty$) row of Eq. (4) are interchanged.

(vi) The most difficult $b = 0$ case is characterized by the oscillatory divergence¹ of the recurrences Eq. (4) in both directions—the mappings $T_n \leftrightarrow T_{n+1}$ have no fixed points in the leading order. Hence, the necessary physical cancellation in Eq. (9) [or boundary conditions in infinity, Eq. (6)] and the corresponding smooth n dependence of T_n must be enforced by the “artificial” asymptotic requirement Eq. (11) compatible with the continuity of both the $b \rightarrow 0^\pm$ limits of the physical $G(E)$ and incompatible with the $b = 0$ conjecture of Ref. 1. We omit the further details here.

¹V. Singh, S. N. Biswas, and K. Datta, Phys. Rev. D **18**, 1901 (1978).

²E. Magyari, Phys. Lett. **81A**, 116 (1981).